Parisa Rabbani

rabbani8@illinois.edu | ParisaRabbani | LadyPary | ladypary.github.io | Scholar

Education

Ph.D. in Computer Science — *08/23 – 05/28 (Expected)*
University of Illinois Urbana-Champaign — *USA*
- GPA: 3.97/4.0
- Specialization: AI Evaluation and Trustworthiness
- Advisor: *Prof. Dilek Hakkani-Tür*

Master of Computer Science — *05/26 (En route to Ph.D.)*
University of Illinois Urbana-Champaign — *USA*

B.S. in Computer Science — *09/19 – 07/23*
University of Tehran — *Iran*
- GPA: 3.77/4.0
- Thesis: *Potential Misuses of Generative Machine Learning Models*
- Advisor: *Prof. Hedieh Sajedi*

Research Interests

- AI Evaluation
- AI Trustworthiness & Safety
- Conversational AI
- Human-AI Collaboration

Publications

- Groundworks: Agents Can Complete Tasks Yet Fail Under Plausible Perturbations In Multi-Turn Interactions
  [*Under Review*] **Parisa Rabbani**, Siyan Li, Yi Evie Zhang, Shubh Jain, Hyeonjeong Ha, and Dilek Hakkani-Tür.
- AgentAbstain: Do LLM Agents Know When Not to Act?
  [*NeurIPS'26 Oral*] Xun Liu, Yi Evie Zhang, Vira Kasprova, **Parisa Rabbani**, Pardis Sadat Zahraei, Tianyu Zhang, Ali Ebrahimpour-Boroojeny, and Varun Chandrasekaran. *Conference on Neural Information Processing Systems, Evaluations & Datasets Track.* Also at COLM 2026 Workshops: Agent Behavior (**spotlight talk**, ~3%) and Lifelong Agents (poster). *Paper* *Website*
- Dialogic Deference: LLMs Can't Judge Who's Right in a Conversation
  [*WiML@NeurIPS'26*] **Parisa Rabbani** and Dilek Hakkani-Tür. *Women in Machine Learning Workshop at NeurIPS.*
- DialDefer: A Framework for Detecting and Mitigating LLM Dialogic Deference
  [*ACL'26 Main Track*] **Parisa Rabbani**, Priyam Sahoo, Ruben Mathew, Aishee Mondal, Harshita Ketharaman, Nimet Beyza Bozdag, and Dilek Hakkani-Tür. *Annual Meeting of the Association for Computational Linguistics.* *Paper* *Code* *Presentation*
- From Fact to Judgment: Investigating the Impact of Task Framing on LLM Conviction in Dialogue Systems
  [*IWSDS'26 Oral*] **Parisa Rabbani**, Nimet Beyza Bozdag, and Dilek Hakkani-Tür. *International Workshop on Spoken Dialogue System Technology.* *Paper* *Code*
- The Social Gaze of LLMs: A Literature Review of Multimodal Approaches to Human Behavior Understanding
  [*Preprint*] Zihan Liu, **Parisa Rabbani**, Veda Duddu, Kyle Fan, Madison Lee, and Yun Huang. *Paper*
- AI-Enhanced Speech-Language Intervention Documentation: Opportunities and Design Goals
  [*AIED'25*] Qingxiao Zheng, Abhinav Choudhry, Zihan Liu, **Parisa Rabbani**, Yuting Hu, Abbie Olszewski, Yun Huang, and Jinjun Xiong. *International Conference on Artificial Intelligence in Education.* *Paper*
- SOAP.AI: A Collaborative Tool for Documenting Human Behavior in Videos through Multimodal Generative AI
  [*CSCW'24 Companion*] Qingxiao Zheng, **Parisa Rabbani**, Yu-Rou Lin, Daan Mansour, and Yun Huang. *ACM Conference on Computer-Supported Cooperative Work and Social Computing.* *Paper* *Demo*

Research Experience

Graduate Research Assistant *Aug 2025 – Present*

Conversational AI Lab, University of Illinois Urbana-Champaign *Champaign, IL, USA*

- Advisor: *Prof. Dilek Hakkani-Tür*
- AI Evaluation: Led *Groundworks* [*Under Review*], an adaptable, end-to-end framework that constructs paired multi-turn evaluations with verifiable target outcomes using plausible perturbations. Across five agents and 1,040 paired evaluations, agents fail on 47% of perturbed scenarios whose originals they complete (averaged across agents): 25% involve targeted hallucination and 22% other failures. Supported by the IARPA BENGAL Cedar project.
- AI Trustworthiness & Safety: Developing *TrustHarness*, an inference-time multi-agent defense for trustworthy agentic AI, funded by Amazon (PI: Prof. Dilek Hakkani-Tür). Introduced *DialDefer* [*ACL'26, WiML'26*] to measure how LLMs shift judgments when evaluating speakers vs. verifying statements, finding a +15.9pp deference shift across 5 LLMs. Studied how task framing affects LLM conviction in dialogue [*IWSDS'26 Oral*].

Research Collaborator *May 2024 – Sep 2024*

AI Institute for Exceptional Education *Urbana, IL, USA*

- Mentor: *Dr. Qingxiao Zheng*
- Human-AI Collaboration: Built *SOAP.AI*, a system for documenting child behavior in therapy videos with GPT-4V and Whisper; co-designed with 17 speech-language pathologists and implemented automatic prompt engineering from clinician preferences. Supported by NSF and U.S. Dept. of Education. [*CSCW'24 Companion, AIED'25*]

Undergraduate Thesis *Fall 2022 – Spring 2023*

University of Tehran *Tehran, Iran*

- Advisor: *Prof. Hedieh Sajedi*
- AI Trustworthiness & Safety: Surveyed misuse risks and societal harms of generative AI (e.g., deepfakes, misinformation). *Thesis*

Selected Projects

LLMs and Lost Translations *Paper*

CS 598: Language, Interfaces, and Communication, UIUC *Spring 2025*

- Evaluated cultural bias inheritance in LLM translation (GPT-4o, Gemini) on 180 CrowS-Pairs sentences in 4 languages via embedding metrics; found bias shifts in up to 90% of gender (Russian) and religion (Korean) outputs.

Deciphering the Digital Zeitgeist *Paper*

CS 598: Computational Social Science, UIUC *Fall 2024*

- Analyzed 68k Reddit comments (SBERT, Poisson regression) on how AI integration affects platform trust; competence distrust fell 26% after the Meta AI launch; tech subreddits stressed competence, non-tech sincerity.

The Many Faces of Computer Science *Paper*

CS 467: Social Visualization, UIUC *Spring 2024*

- Visualized CS faculty diversity via ML-generated facial composites (3,900+ images, 48 U.S. universities); found 67% masculine representation, with top-ranked schools skewing white male and minority-serving ones mirroring student diversity.

Professional Service

Organizer: Midwest Speech and Language Days (MSLD) 2026

Program Committee: SIGDIAL 2026

Reviewer: NeurIPS 2026; NeurIPS 2026 (IAEval Workshop); ACL Rolling Review 2026 (January ARR); NeurIPS 2025 (MTI-LLM Workshop); ACM CHI 2025 (Late-Breaking Work)

Presentations

Agentic Evaluation Should Consider the "Imperfect" Scenarios - *Slides*

Invited Talk. ACL 2026 Birds-of-a-Feather: *What Makes an Enterprise Agent Benchmark Useful?*

Improving Hallucination Detection in Dialog via Social Framing Analysis - *Poster*

Poster. Midwest Speech and Language Days 2026

DialDefer: Detecting Dialogic Deference in LLM Judges - *Poster*

Invited Poster. Amazon Trusted AI Symposium 2026. **Poster.** MLSS 2026, Columbia University.

Conversational Judgment Sycophancy - *Poster*

Poster. Amazon-UIUC AICE Fall Symposium 2025.

Designing for Human Behavior Analysis using Multi-Modal AI

Oral. AI Institute for Exceptional Education Workshop 2024.

FMC-LIFE: Smart Games for Autism & Alzheimer's

Oral. RoboCup Iran Open 2017. **Best presentation award.**

Technical Skills

Languages:	Python, C++, SQL, LaTeX
NLP & LLMs:	Fine-Tuning, Prompt Engineering, LLM Evaluation & Benchmarking, LLM Judge Pipelines, Red-Teaming, Empirical Study Design
ML & Data:	PyTorch, Scikit-learn, Pandas, XGBoost, Regression, Classification
HCI & Research:	Needfinding, Prototyping, User Studies, Survey Design, Thematic Analysis

Teaching Experience

Graduate Teaching Assistant

University of Illinois Urbana-Champaign — *Fall 2024 – Spring 2026*

- CS 568 User-Centered Machine Learning, *Prof. Ranjitha Kumar (Spring 2026)* — grading.
- CS 465 User Interface Design, *Prof. Brian Bailey (Fall 2024, Fall 2025)* — studio sessions and project feedback.
- CS 416 Data Visualization, *Prof. John Hart (Summer 2025)* — office hours and grading.
- CS 101 Intro to Programming, *Dr. Liia Butler & Dr. Nico Ritschel (Spring 2025)* — lab sessions and exam reviews.

Awards & Selected Programs

Machine Learning Summer School (MLSS) 2026. *Columbia University & Bloomberg (June 2026)*

Accepted to attend (~16% acceptance rate); two-week research program on LLM evaluation, safety, and agentic AI.

National Scholarship & Academic Honors. *University of Tehran (2019–2023)*

Ranked top 1% among 164,000 students in the national entrance exam (Konkur); awarded full tuition waiver, foundation scholarship, and excellence title 5 of 6 semesters.

FMC-LIFE: Smart Games for Autism & Alzheimer's. *RoboCup Iran Open & NODET (2017–2018)*

FMC-LIFE project: 1st place + Best Teamwork + Best Presentation (RoboCup 2017); Top Tech Project (NODET 2018).

Volunteer Activities

Girls Who Code Spotlight Speaker — *Slides | Spring 2025*

- Delivered HCI research presentation to 10–20 middle school students.

CS Visit Days Graduate Ambassador — *Slides | Spring 2024, Spring 2025*

- Facilitated Diversity @ Illinois sessions and virtual Q&As for prospective international PhD students.